Cook Inlet Beluga Readme File

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Workflow

To generate quantitative metabarcoding we are integrating multiple datastreams:

- **metabarcoding of environmental DNA** samples using MiFish 12S primers,
- metabarcoding from **mock communities** with known starting input concentrations using MiFish 12S primers,
- and **qPCR of Coho salmon** (*Oncorhynchus kisutch*)

[**Note**: qPCR was run on 3 other species Beluga (*Delphinapterus leucas*), Eulachon (*Thaleichthys pacificus*), and King salmon (*Oncorhynchus tshawytscha*).

Here we follow the workflow laid out in Andruszkiewicz Allan et al. 2023 (Allan et al. 2023).

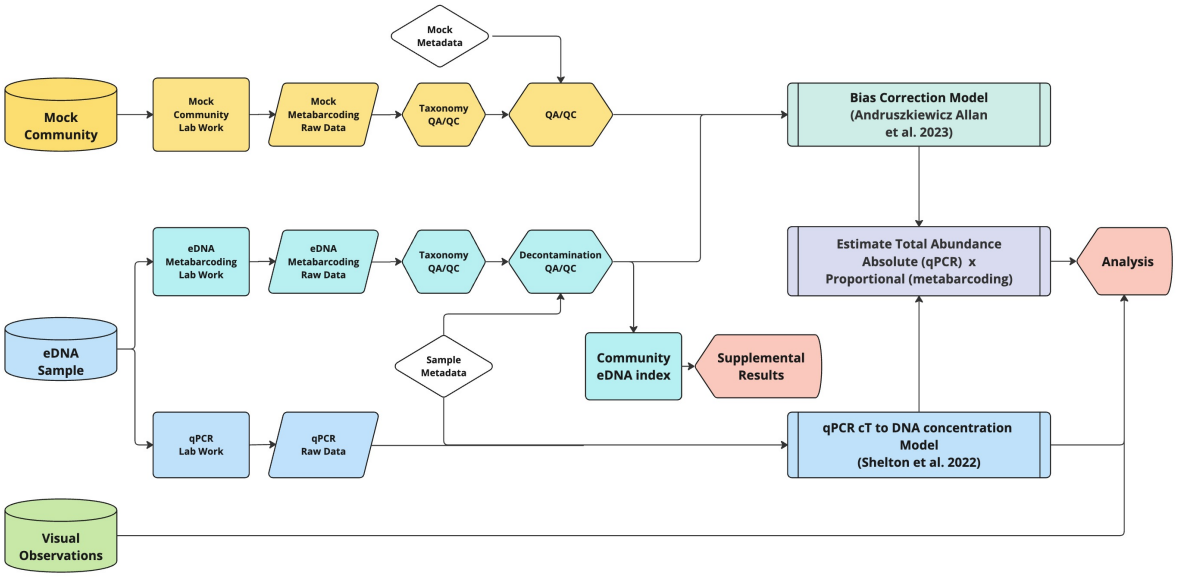


Figure 1: image

Figure 2. Workflow diagram of quantitative metabarcoding approach.

Data Organization & QA/QC

As illustrated in Figure 2, there are a series of data organization and QA/QC steps needed to process the multiple data streams and files into a cohesive analysis. This section outlines the processes in more granular detail.

Mock community data

- The raw and processed data has been published in Shelton et al. 2023 (Shelton et al. 2023) and Gold et al. 2023(Gold et al. 2023).
- Mock community metabarcoding data was processed using the Anacapa Toolkit pipeline following Gold et al. 2021 with an additional taxonomy fix step implemented for salmon (**Oncorhynchus sp.**) following Shelton et al. 2023.
- Data was post processed to remove poorly sequenced mock communities and confirm annotations to known input species following Shelton et al. 2023.

Input Data

File	Description	Type	Path
`Zack_MockCommunity/`	Raw fastq reads of mock community data	Data	`./data/Mock_community_data/data/mock_sequences/Zack_MockCommunity_raw_sequences/`
`Anacapa_output_updated_20230106/`	Out processed data from Anacapa Toolkit as described in Gold et al. 2023	Data	`./data/Mock_community_data/data/mock_sequences/Anacapa_output_updated_20230106/`
`input_dna_conc_communities_20210103.csv`	Sample metadata for mock communities including the known input DNA concentrations for each sample	Data	`./data/Mock_community_data/data/input_dna_conc_communities_20210103.csv`
`hash.key_updated.RDS`	Updated taxonomic assignments for ASVs with GenBank Accessions and full taxonomic paths. This includes improved *Oncorhynchus* assignments from `taxonomy_matcher_12S_20210106.Rmd`	Data	`./data/Mock_community_data/data/mock_sequences/hash.key_updated.RDS`

Code

File	Description	Type	Path
`mc31_organization_20261109.Rmd`	This file conducts QA/QC and organizes the mock community metabarcoding data, bringing together observed sequences reads and known input concentrations.	Code	`./data/Mock_community_data/code/mc31_organization_20261109.Rmd`

File	Description	Type	Path
taxonomy_matcher_12S_20210106.Rmd	This file improves taxonomic assignments made to ASVs in the genus *Oncorhynchus* through a simple paired BLAST + phylogenetic analysis of ASVs and known reference sequences as described in Gold et al. 2023. Importantly we note that this specifically ignores GenBank Accessions NC_002980 (*Oncorhynchus tshawytscha*) and NC_017838 (*Oncorhynchus keta*). Both sequences are not regionally relevant (from Russia) and appear to be potential misannotations or hybridization events.	Code	./data/Mock_community_data/code/taxonomy_matcher_12S_20210106.Rmd
run-anacapa_qc_mc31_20220104.sh	Anacapa QA/QC ASV generation script run using the California Current Large Marine Ecosystem specific FishCard reference database described in Gold et al. 2021	Code	./data/Mock_community_data/code/run-anacapa_qc_mc31_20220104.sh
run-anacapa_qc_mc31_20220105.sh	Anacapa QA/QC ASV generation run script run using the Global FishCard reference database described in Gold et al. 2021	Code	./data/Mock_community_data/code/run-anacapa_qc_mc31_20220105.sh
run-anacapa-classifier-mc31_20210105.sh	Anacapa classifier script run using the California Current Large Marine Ecosystem specific FishCard reference database described in Gold et al. 2021	Code	./data/Mock_community_data/code/run-anacapa-classifier-mc31_20210105.sh
run-anacapa-classifier-mc31_combo_20210105.sh	Anacapa classifier script run using the Global FishCard reference database described in Gold et al. 2021	Code	./data/Mock_community_data/code/run-anacapa-classifier-combo_20210105.sh

Output Data

File	Description	Type	Path
mifish_mock_community_data.RDS	Output file of organized mock community metabarcoding data. This data is used to calibrate observations from eDNA metabarcoding.	Data	./data/Mock_community_data/data/mifish_mock_community_data.RDS

Metabarcoding Environmental data

- Enviromental DNA Metabarcoding data was processed using the Anacapa Toolkit pipeline following Gold et al. 2021 (Gold et al. 2021) with an additional taxonomy fix step implemented for salmon (*Oncorhynchus* sp.) following Shelton et al. 2023. A total of five sequencing runs were conducted for this project, 3 runs from Kenai River samples and 2 runs from Twentymile River samples.
- Resulting amplicon sequence variant tables (ASVs) were then further processed using the Gruinard decontamination pipeline as implemented in Gold et al. 2022 (Gold et al. 2022). Scripts to run the decontamination are in the decontam/ directories located within each of the five sequencing run directories.
- Metadata was organized and cleaned to allow for downstream sample merging.

- eDNA index was derived for all species following Kelly et al. 2019 (Kelly et al. 2019). This was done as high quality mock community data was only available for a distinct subset of taxa and this allowed for community analyses across a broader swath of observed taxa for supplemental analyses.

Data

File	Description	Type	Path
`Anacapa_processed_sequence_data/`	Data processed from 5 sequencing runs (`Kenai_1`, `Kenai_2`, `Kenai_3`, `Twentymile_1`, `Twentymile_2`) using the Anacapa Toolkit as described in Gold et al. 2021.	Data	`./data/eDNA_data/data/Anacapa_processed_sequence_data/`
`Raw_sequence_data`	Raw fastq reads	Data	`./data/eDNA_data/data/Raw_sequence_data/`

Code

File	Description	Type	Path
`Anacapa_scripts/`	Directory of Anacapa QA/QC ASV generation & Anacapa classifier scripts run using using the `Global FishCard reference database` described in Gold et al. 2021	Code	`./data/eDNA_data/code/Anacapa_scripts/`
`taxonomy_fix/`	Directory of scripts and data used to correct taxonomy. This approach is identical to that described for the mock community above.	Code & Data	`./data/eDNA_data/code/taxonomy_fix/scripts/`
`decontam/`	These subdirectorties within each of the five sequencing run directories includes the code to run decontamination of each output Anacapa ASV table. Decontamination was completed with the following steps: 0) Retain only paired merged reads , 1) Account for estimated rate tag jumping (index hopping) by subtracting proportional read abundances observed in positive controls from samples, 2) Retain PCR technical replicates with greater than 30K reads, 3) Retain ASVs with a probability of detection greater than the maximum estimated probability of an ASV being detected only once. This was implemented using a Site Occupancy Modeling framework on the pattern of presence of ASVs observed across the data set, 4) Discard Biological and PCR Technical replicates with greater than 99% dissimilarity from all other samples.	Data	`.data/eDNA_data/data/Anacapa_processed_sequence_data/[insert river]/[insert sequencing run] /decontam`

Outputs

File	Description	Type	Path
pre_occupancy_ results_ sum.taxonomy _tech_reps_ separate_ read_counts.RDS	Long formatted data table of final ASVs and assigned taxonomy with relevant sample metadata from eDNA samples.	Data	./data/Anacapa_processed _sequence_data/ [kenai/twetnymile]/ */decontam/ Output_R/pre_occupancy _results_sum.taxonomy _tech_reps_ separate_read_ counts.RDS

qPCR Data

- qPCR data was processed following Shelton et al. 2022 (Shelton et al. 2022) using a bayesian model to estimate qPCR concentrations of environmental data from known standards.
- Metadata was organized and cleaned to allow for downstream sample merging.

Data

File	Description	Type	Path
qPCR_raw/	Raw qPCR results from 10 distinct plates. Eulachon, Coho, and Chinook were multiplexed together. Beluga was run separately. Each plate includes it's own standard curve with known concentrations as well as samples with unknown concentrations.	Data	.data/qPCR_results/ data/qPCR_raw/
sample_metadata/	Sample metadata from each of the 10 qPCR plates	Data	./data/qPCR_results/ data/sample_metadata/

Code

File	Description	Type	Path
qpcr_formatting _20220624.Rmd/	Code to organize, merge, and clean up the raw qPCR and sample metadata. The resulting output is qPCR_AK_data.RDS which is a merged long data table with all results needed for downstream analyses	Code	./data/qPCR_results/ code/qpcr_formatting _20220624.Rmd
qPCR_attempt_ 20230223.Rmd	Code modified from Andruszkiewicz Allan et al. 2023 to import qPCR data and run a STAN based model to estimate concentrations of DNA based on standards using the model and code developed originally by Ole Shelton used in Shelton et al. 20	Code	.data/qPCR_results/ code/qPCR_attempt_ 20230223.Rmd

Outputs

File	Description	Type	Path
qPCR_AK_data.RDS	Long formatted data table of qPCR data including both observed values and standards with relevant sample metadata.	Data	.data/qPCR_results/qPCR_AK_d
[species]_input_csv_for_stan.csv	Formatted data for input into STAN model to estimate true DNA concentrations.	Data	.data/qPCR_results/data/qPCR_results/[species]_input_csv_for_stan.csv
qMod_[species].RDS	Full STAN output for beluga, chinook, coho, and eulachon DNA concentrations.	Data	.data/qPCR_results/data/qPCR_results/qMOD_[species].RDS
qMod_[species]_tibble.RDS	Full STAN output for beluga, chinook, coho, and eulachon DNA concentrations converted to a tibble format for ease of use.	Data	.data/qPCR_results/data/qPCR_results/qMOD_[species]_tibble.RDS

Visual Beluga Observations data

1. Alaska Beluga Monitoring Program community science derived visual data. Observations are derived from 2 hour visual observations centered around high tide.
2. Sonia Kumar visual observations (following similar methods with the exception of not stricly being associated with high tides).

Data

File	Description	Type	Path
AKBMP-Fall2021-eDNA_20220208.csv	AKBMP Visual Observations of Belugas from Fall 2021	Data	./data/Visual_observations/data/AKBMP-Fall2021-eDNA_20220208.csv
AKBMP-KenaiRiver-Sightings_spring.csv	AKBMP Visual Observations of Belugas on Kenai River	Data	./data/Visual_observations/data/AKBMP-KenaiRiver-Sightings_spring.csv
sonia_kumar_visual_observations.csv	Sonia Kumar's Visual Observations of Belugas on Kenai River	Data	./data/Visual_observations/data/sonia_kumar_visual_observations.csv

Code

File	Description	Type	Path
beluga_visual_observation_20240823.Rmd	Code to organize, merge, and clean up the visual observations. The resulting output is beluga_visual_3.RDS which is a merged long data table with all results needed for downstream analyses	Code	./data/Visual_observations/code/beluga_visual_observation_20240823.Rmd

Outputs

File	Description	Type	Path
beluga_visual_3.RDS	Long formatted data table of visual beluga observations data	Data	./data/Visual_observations/d

Bias Corrected Metabarcoding

- Mock communities were used to calibrate eDNA metabarcoding results following Shelton et al. 2023. Here we focus on only the five common *Oncorhynchus* salmon species: Coho, King, Chum, Pink, and Sockeye. All five species had sufficient mock community data to accurately estimate amplification efficiencies.

Data

File	Description	Type	Path
Metabarcoding Data	Decontaminated metabarcoding data for 5 runs	Data	./data/eDNA_data/ data/Anacapa_processed_ sequence_data/[river]/ [sequencing run]/decontam/ Output_R/pre_occupancy_ results_sum.taxonomy_ tech_reps_separate_ read_counts.RDS
Sample Metadata	Sample metadata for 5 runs	Data	./data/Metadata/
mifish_mock_ community_data.RDS	This output file of organized mock community metabarcoding data used to calibrate observations from eDNA metabarcoding.	Data	./data/Mock_community_ data/data/mifish_mock_ community_data.RDS

Code

File	Description	Type	Path
combined_beluga_ metabarcoding_data_ 20240824.Rmd	Code to organize and merge eDNa metabarcoding data. The resulting output is combined_data_long_reads_c.RDS which is a merged long data table with all results needed for downstream analyses	Code	./analysis/scripts/ combined_beluga_ metabarcoding_data_ 20240824.Rmd
qmeta_model_ 20231213.R	Code to run metabarcoding bias correction. Code modified from Andruszkiewicz Allan et al. 2023. Important output is 20231210_bayesout_salmon_only.RDS	Code	./analysis/scripts/ qmeta_model_20231213.R

Outputs

File	Description	Type	Path
20231210_bayesout_salmon_only.RDS	Model output of bias corrected metabarcoding data for 5 species of salmon.	Data	./analysis/Output/20231210_bayesout_salmon_only.RDS

Estimating Abundance

Metabarcoding Proportions x qPCR Absolute Abundance

- Proportional metabarcoding data for Coho was cross multiplied against qPCR absolute abundance concentrations to derive the total number of DNA molecules in a given sample following Andruszkiewicz Allan et al. 2023. Coho was chosen as the anchor species as it was the species amplified via qPCR that was the most commonly detected species across the metabarcoding data set. Estimates of quantitative abundance for the five species with bias corrected metabarcoding proportional data were then derived cross multiplying the observed proportional reads with the estimated total DNA molecules per sample.

Data

File	Description	Type	Path
20231210_bayesout_salmon_only.RDS	Model output of bias corrected metabarcoding data.	Data	./analysis/Output/20231210_bayesout_salmon_only.RDS
qMod_[species].RDS	Full STAN output for beluga, chinook, coho, and eulachon.	Data	.data/qPCR_results/data/qPCR_results/qMOD_[species].RDS
qMod_[species]_tibble.RDS	Full STAN output for beluga, chinook, coho, and eulachon converted to a tibble format for ease of use.	Data	.data/qPCR_results/data/qPCR_results/qMOD_[species].RDS
qPCR_AK_data.RDS	Long formatted data table of qPCR data including both observed values and standards with relevant sample metadata.	Data	.data/qPCR_results/qPCR_AK_data.RDS

Code

File	Description	Type	Path
qPCR_qm_merge_attempt_20231213.R	Code to estimate abundance by multpying bias corrected proportions x qPCR concentrations. Code modified from Andruszkiewicz Allan et al. 2023.	Code	./analysis/scripts/qPCR_qm_merge_attempt_20231213.R

Outputs

File	Description	Type	Path
coho_anchor_predicted_DNA_conc.RDS	Model output using Coho as anchor for estimated abundance.	Data	./data/Visual_observations/data/beluga_visual_3.RDS

File	Description	Type	Path
coho_anchor_abs_abundance_notflowcorrected.RDS	Model output using Coho as anchor for estimated abundance.	Data	./data/Visual_observations/data/beluga_visual_3.RDS

Analyses

Analyses conducted for the manuscript

Main Manuscript

File	Description	Type	Path
CIB_analysis_trimmed_20261109.Rmd	Main Analysis Document	Code	analysis/scripts/CIB_analysis_trimmed_20261109.Rmd

- Beluga occurrence. Visual, qPCR, and metabarcoding detection rates for beluga were compared before, during, and after beluga river habitat use.
- Twentymile river quantitative metabarcoding time series. Estimated abundance of fish species from eDNA were compared before and during observations of Cook Inlet Beluga in Twentymile river.
- Summary of general statistics and diversity observed via eDNA metabarcoding data.
- General statistics and figures of fish species occurrences during periods of beluga observation on both Kenai and Twentymile River.
- Temporal and Spatial Variation. Specifically, an analysis of factors explaining variance in Kenai River environmental data, examining fish communities across time, site, and depth to inform future sampling strategies.
- Non-detections and the effects of swamping and rare molecules.

Supplemental

File	Description	Type	Path
Supplemental_Analsyses_20240825.Rmd	Supplemental Analysis Document for Appendix 5	Code	analysis/scripts/Supplemental_Analsyses_20240825.Rmd

- Appendix 5 qPCR Anchor Comparisons

References

Allan, Elizabeth Andruszkiewicz, Ryan P Kelly, Erin R D'Agnese, et al. 2023. "Quantifying Impacts of an Environmental Intervention Using Environmental DNA." *Ecological Applications* 33 (8): e2914.

Gold, Zachary, Emily E Curd, Kelly D Goodwin, et al. 2021. "Improving Metabarcoding Taxonomic Assignment: A Case Study of Fishes in a Large Marine Ecosystem." *Molecular Ecology Resources* 21 (7): 2546–64.

Gold, Zachary, Andrew Olaf Shelton, Helen R Casendino, et al. 2023. "Signal and Noise in Metabarcoding Data." *Plos One* 18 (5): e0285674.

Gold, Zachary, Adam R Wall, Teia M Schweizer, et al. 2022. "A Manager's Guide to Using eDNA Metabarcoding in Marine Ecosystems." *PeerJ* 10: e14071.

Kelly, Ryan P, Andrew Olaf Shelton, and Ramon Gallego. 2019. "Understanding PCR Processes to Draw Meaningful Conclusions from Environmental DNA Studies." *Scientific Reports* 9 (1): 12133.

Shelton, Andrew Olaf, Zachary J Gold, Alexander J Jensen, et al. 2023. "Toward Quantitative Metabarcoding." *Ecology* 104 (2): e3906.

Shelton, Andrew Olaf, Ana Ramon-Laca, Abigail Wells, et al. 2022. "Environmental DNA Provides Quantitative Estimates of Pacific Hake Abundance and Distribution in the Open Ocean." *Proceedings of the Royal Society B* 289 (1971): 20212613.